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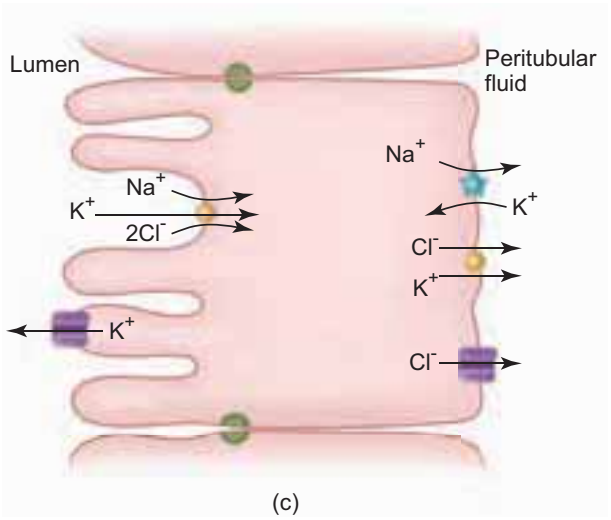


Figure 8.9 (c) Thick ascending limb

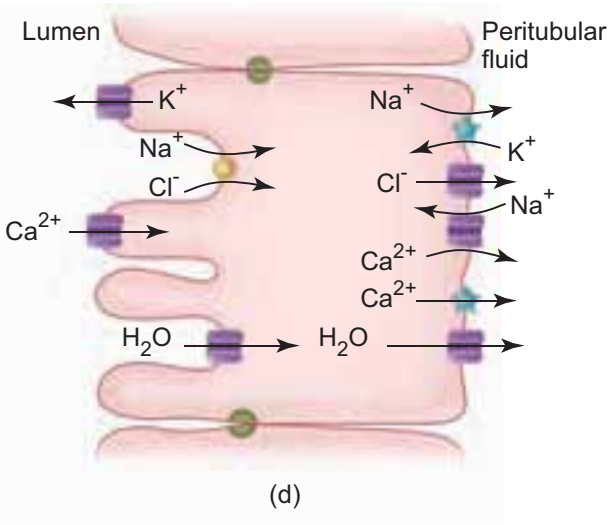


Figure 8.9 (d) Transport in the distal tubule

the filtrate of the DCT (Figure 8.9(d)). Most of the reabsorption from this point is dependent on the body's need and is regulated by hormones. Reabsorption of bicarbonate (HCO_3^2) takes place to regulate the blood pH. Homeostasis of K^1 and Na^1 in the blood is also regulated in this region.

Aquaporins are water–permeable channels (membrane transport proteins) that allow water to move across the epithelial cells in relation to the osmotic difference from the lumen to the interstitial fluid.

Collecting duct is permeable to water, secretes K^1 (potassium ions are actively transported into the tubule) and reabsorbs Na^1 to produce concentrated urine. The change in permeability to water is due to the presence of number of water-permeable channels called **aquaporins**.

Tubular secretion- Substances such as H^1, K^1, NH_4^1, creatinine and organic acids move into the filtrate from the peritubular capillaries into the tubular fluid. Most of the water is absorbed in the proximal convoluted tubule and Na^1 is exchanged for water in the loop of Henle. Hypotonic fluid enters the distal convoluted tubule and substances such as urea and salts pass from peritubular blood into the cells of DCT. The urine excreted contains both filtered and secreted substances. Once it enters the collecting duct, water is absorbed and concentrated hypertonic urine is formed. For every H^1 secreted into the tubular filtrate, a Na^1 is absorbed by the tubular cell. The H^1 secreted combines with HCO_3^2, HPO_3^2 and NH_3^2 and gets fixed as $H_2CO_4^1$, $H_2PO_4^1$ and NH_4^1 respectively. Since H^1 gets fixed in the fluid, reabsorption of H^1 is prevented.

Osmolarity - The solute concentration of a solution of water is known as the solutions osmolarity, expressed as milliosmoles /liter (mOsm/L)

Formation of concentrated urine
Formation of concentrated urine is accomplished by kidneys using counter